

Jiayi (Emma) Xu

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Education

University of Toronto (UTSG) Toronto, Canada (Sep 2025 - Sep 2026)

Master in Physics

GPA: 3.9/4.0

Research on Filter Design for Balloon Borne Telescope and DMD based Multi-Object Spectrograph

University of Toronto (UTSG) Toronto, Canada (Sep 2021 - Jun 2025)

Honours BSc in Astronomy and Physics with High Distinction

Advanced courses GPA (upper year math, physics, astronomy): 3.83/4.0

Cumulative GPA: 3.69/4.0

Relevant courses: Relativistic Theory, Quantum Mechanics, Electromagnetism, Classical Mechanics, Foundation of Astronomy and Astrophysics, Time Series Analysis

Awards/Scholarships/Fellowships

University of Toronto Fellowship (UTF)(\$24,658 CAD) (2025 - 2026)

University of Toronto, School of Graduate Study fellowship for master degree students in Physics

SURP(\$10,880 CAD) (May-Sep 2025)

2025 University of Toronto Summer Undergraduate Research Program in Astronomy & Astrophysics

NSERC-USRA (\$7,500 CAD) (May-Sep 2024)

Natural Sciences and Engineering Research Council of Canada – Undergraduate Student Research Award

The Roy Alvin Hope Scholarship (\$1,000 CAD) (2023 - 2024)

For third-year students with a Cumulative GPA above 3.5/4.0

Regents in-course Scholarship (\$1,000 CAD) (2022 - 2023)

For second-year students with a Cumulative GPA above 3.5/4.0

Research Experience

Filter Design for Giga-pixel Balloon Borne Imaging Telescope (GigaBIT)
(Sep 2025 - Sep 2026)

- Designed optimized band-pass filters tailored for galactic archaeology and stellar evolution studies, successfully resolving parameter degeneracies in multiple stellar populations (MSPs) within globular clusters and filtering out field star contamination for dim targets.
- Trained advanced bandpass-color empirical models using sequential machine learning methods; implemented a 4D Fisher Information Matrix framework to systematically suppress parameter errors by accounting for complex multi-parameter covariances and correlations (T_{eff} , $\log g$, $[Fe/H]$, $[\alpha/Fe]$).
- Developed a dynamic observation time allocation optimizer based on Fisher analysis derivatives, transforming scientific parameter precision constraints into optimized multi-filter exposure management strategies.

Digital Micro-mirror Device based Multi-Object Spectrograph (DMD-MOS) calibration and testing

(May 2025 -Sep 2026)

- Led laboratory testing and instrumentation integration; performed optical alignment, wavelength calibration, and image quality assessments using custom Python routines, including computing point spread function (PSF) full-width at half-maximum (FWHM) via Gaussian fitting.
- Co-developed an automated spectral collection algorithm utilizing a homography matrix to map target detector coordinates back onto the DMD micro-mirror plane, enabling precision slit positioning.
- Executed on-sky testing and field observations to rigorously evaluate overall system reliability under real-observation conditions.

Lattice Simulations of Gravitational Waves During Inflation

(Sep 2024 – Sep 2025)

- Simulating Stochastic Gravitational Wave Background (SGWB) emitted during inflation. Combining theory studies on Relativity, early universe evolution; computational simulation with Fortran; and data visualization with Python.
- Utilizing lattice simulations to explore the non-linear dynamics of the early universe, focusing on α -attractor models of inflation, to determine their gravitational wave background signatures.
- Developing independent learning abilities on theoretical studying and programming by modify models and initial conditions of the lattice simulation. Aiming to constraining the inflationary model.

Gemini Infrared Multi-Object Spectrography Diffraction Grating Evaluations

(Sep 2024 – May 2025)

- Collaborated with cross-disciplinary teams to optimize the GIRMOS diffraction grating system. Combined optical engineering, statistical analysis (Python) and instrumental research technics.
- Developed hands-on experiment skills by setting up the system; and improved the application ability from theory to practical by testing photon efficiency and wavefront error of the grating units with photodetectors.

Professional Skills

- **Python:** Proficient in data modeling, data visualization and large datasets processing.
- **Modelling:** Expertise in statistical modeling, lattice modeling, bias correction, data optimization, and Monte Carlo analysis.
- **Experiment design:** Skilled in designing experiments to test the optical performance, collecting and analyzing spectra and imaging data.
- **Large-scale instruments:** Experience with assembly, setup, operation, and maintenance of optical systems.
- **Microsoft Office:** Proficient in Word, PowerPoint, Excel for data analysis and creating presentations.
- **LaTeX:** Experience in reports-writing, formatting complex formulas and tables.

Research Interests

Cosmology Early universe evolution, simulations on inflationary models. High energy physics and the evolution of energetic particles after the Big Bang. Fate of the universe, prediction of the future trend of the universe.

Instrumental Research Experimental-based research. Telescope design, building, calibration, testing, examination. Optics related projects.

Additional Research Experience

NANO Optical Particle Applications (May-Sep 2024)

- Collaborated on nano-particle optical applications on window models, blocking specific wavelengths with low-emissivity coatings of silver and aluminum nitride.
- Mastered advanced techniques in using testing instruments; led tests on heat/corrosion resistance with data analyzed via Fermi-Dirac functions and Monte Carlo methods.

Statistical climate downscaling study (PHY371) (Sep 2023 - Apr 2024)

- Simulated 50-year climate change of Arctic region using multi-model ensemble simulations “UTCDW” (University of Toronto Climate Downscaling Workflow) in Python.
- Optimized the model resolution using statistical techniques, ensuring the robust and resistance properties of the models.

Practical astronomy course (AST326) (Sep 2023 - Apr 2024)

- Collected, analyzed and plotted spectra of a Type Ia supernova and an exoplanet, both captured using CCD astronomical cameras.
- Simulated asteroid orbits using the NASA HORIZONS database, calculated the key orbital parameters of the asteroid, and predicted the future position and route.

Volunteer Experiences

Skate Canada International 2024 (Oct 2024)

- Volunteered as a backstage security guard of a global broadcast international skating competition.

TCS Toronto waterfront marathon (Oct 2024)

- Volunteered as a marshal in global marathon event run by TATA Consultancy Services (TCS).

UofT AstroTour (2022 - 2024)

- Monthly popular science activity organized by UofT students. Introduced 3D-printed spaceship/telescope models to audience.